AQ8.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

Consider an ordered basis $\gamma = \{b_1, b_2, b_3\}$γ = {b1, b2, b3} of the inner product space R^3 R3 with respect to the dot product, where $b_1 = (1, 2, 3)$, $b_2 = (1, 1, 1)$b1 = (1, 2, 3), b2 = (1, 1, 1) and $b_3 = (0, 2, 1)$b3 = (0, 2, 1). Let $\beta = \{a_1, a_2, a_3\}$β = {a1, a2, a3} be the orthonormal basis which is obtained using the Gram-Schmidt process from the basis γγ. Use this information to answer the questions 1 and 2.

1 point

The vector a_2 a2 is

- ○ $\left(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$(21 √ 4, 21 √ 1, 21 √ 2)
- ○ $\left(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{-2}{\sqrt{21}}\right)$(21 √ 4, 21 √ 1, 21 √ −2)
- ○ $\left(\frac{-4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$(21 √ −4, 21 √ 1, 21 √ 2)
- ○ $\left(\frac{4}{\sqrt{21}}, \frac{-1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$(21 √ 4, 21 √ −1, 21 √ 2)

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{-2}{\sqrt{21}})$(21
√
4, 21
√
1, 21
√
−2)

1 point

The vector a_3 a3 is

○ $\frac{1}{\sqrt{6}}(-1, 2, -1)$ 6
√
1(−1, 2, −1)

○ $\frac{1}{\sqrt{6}}(-1, -2, -1)$ 6
√
1(−1, −2, −1)

○ $\frac{1}{\sqrt{6}}(1, 2, -1)$ 6
√
1(1, 2, −1)

○ $\frac{1}{\sqrt{6}}(1, 2, 1)$ 6
√
1(1, 2, 1)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{1}{\sqrt{6}}(-1, 2, -1)$ 6
√
1(−1, 2, −1)

Level 2

1 point

Let $v_1 = (1, 0, 1, 1)$v1 = (1, 0, 1, 1) and $v_2 = (0, 1, 1, 1)$v2 = (0, 1, 1, 1) be the vectors from the inner product space R^4 R4 with respect to the dot product. If $v_3 = v_2 + av_1$ v3 = v2 + av1 where $a \in R$ a ∈ R and v_1, v_3 v1, v3 are orthogonal, then

○ $a = -2/3$a = −2/3

○ $a = 2/3$a = 2/3

○ $a = 1/3$a = 1/3

○ $a = -1/3$a = −1/3

No, the answer is incorrect.
Score: 0

Accepted Answers:
$a = -2/3$a = −2/3

1 point
Let WW be a subspace of the inner product space R^4 R4 with respect to dot product and $\{v_1, v_2, v_3\}$ {v1, v2, v3} be an ordered basis of WW, where $v_1 = (1,1,1,1), v_2 = (1,1,1,0), v_3 = (1,1,0,0)$v1 = (1, 1, 1, 1), v2 = (1, 1, 1, 0), v3 = (1, 1, 0, 0). Which of these is the orthonormal basis of WW obtained using the Gram-Schmidt process?

○ $\{\frac{1}{2}(1,1,1,1), \frac{1}{2\sqrt{3}}(1,1,1,-3), \frac{1}{\sqrt{6}}(1,1,-2,0)\}${21(1, 1, 1, 1), 2 3 √ 1(1, 1, 1, −3), 6 √ 1(1, 1, −2, 0)}

○ $\{\frac{1}{2}(1,1,1,1), \frac{1}{2\sqrt{3}}(1,1,1,-3), \frac{1}{\sqrt{6}}(1,1,2,0)\}${21(1, 1, 1, 1), 2 3 √ 1(1, 1, 1, −3), 6 √ 1(1, 1, 2, 0)}

○ $\{\frac{1}{2}(1,1,1,1), \frac{1}{2\sqrt{3}}(1,1,1,-3), \frac{1}{\sqrt{6}}(3,1,-2,0)\}${21(1, 1, 1, 1), 2 3 √ 1(1, 1, 1, −3), 6 √ 1(3, 1, −2, 0)}

○ $\{\frac{1}{2}(1,1,1,1), \frac{1}{2\sqrt{3}}(1,1,1,-3), \frac{1}{\sqrt{6}}(3,1,2,-2)\}${21(1, 1, 1, 1), 2 3 √ 1(1, 1, 1, −3), 6 √ 1(3, 1, 2, −2)}

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{\frac{1}{2}(1,1,1,1), \frac{1}{2\sqrt{3}}(1,1,1,-3), \frac{1}{\sqrt{6}}(1,1,-2,0)\}${21(1, 1, 1, 1), 2 3 √ 1(1, 1, 1, −3), 6 √ 1(1, 1, −2, 0)}

1 point
Let $a = (\frac{2}{3}, \frac{2}{3}, \frac{1}{3})$a = (32, 32, 31) be a vector from the inner product space R^3 R3 with respect to dot product and $W = \{(x,y,z) \in R^3 \mid \langle (x,y,z), (\frac{2}{3}, \frac{2}{3}, \frac{1}{3}) \rangle = 0\}$W = {(x, y, z) ∈ R3 ∣ ⟨(x, y, z), (32, 32, 31)⟩ = 0} be a subspace of R^3 R3. Then which of the following is (are) a basis of WW?

☐ $\{(1,0,2), (0,1,2)\}${(1, 0, 2), (0, 1, 2)}

☐ $\{(1,0,-2), (0,1,-2)\}${(1, 0, −2), (0, 1, −2)}

☐ $\{(-1,0,2), (0,-1,2)\}${(−1, 0, 2), (0, −1, 2)}

- [ ] $\{(1,0,2),(0,1,-2)\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(1,0,-2),(0,1,-2)\}$
$\{(-1,0,2),(0,-1,2)\}$

Check Answers
Your score is: 0/5